

UC MATH 600 MATHEMATICS FOR COMPUTER SCIENCE · MODULE 02 · LECTURE NOTES

Logic and Proof Technique

Three techniques that cover almost everything.

MODULE OUTCOMES

- Read and write quantified statements precisely.
- Negate a statement correctly.
- Write direct and contrapositive proofs.
- Write proofs by contradiction, and know when not to.
- Choose a technique from the shape of the statement.

1 Statements

ALMOST EVERY STATEMENT IN THIS PROGRAM HAS THE FORM 'FOR ALL x , IF $P(x)$ THEN $Q(x)$ '

'Every even integer greater than two is the sum of two primes'; 'if a graph is connected then it has a spanning tree'; 'if the input is sorted then binary search terminates in $O(\log n)$ steps' — all the same shape, and the shape tells you how to start (§4).

The order of the quantifiers matters absolutely: 'for every x there is a y such that $R(x,y)$ ' and 'there is a y such that for every x , $R(x,y)$ ' are different statements, and the second is much stronger — it demands one y that works for all x at once.

Which is the commonest source of a wrong proof in this course — proving the weak one and claiming the strong one — and it happens because English hides the distinction: 'everyone has a favourite colour' and 'there is a colour everyone favours' are both natural readings of a careless sentence.

So write the quantifiers out, in order, before you start — which costs one line and settles what you are actually trying to prove, and which Project 1 asks you to do at the top of every proof.

THE RULES

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not (for all x, P(x))      = exists x, not P(x)
not (exists x, P(x))      = for all x, not P(x)
not (P and Q)             = not P or not Q
not (P or Q)              = not P and not Q
not (if P then Q)         = P and not Q
<-- this one surprises people

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SO TO NEGATE

walk the quantifiers left to right, flipping each one, and negate the innermost statement.

EXAMPLE

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"for every  $\epsilon > 0$  there is an  $N$  such that for all
 $n > N$ ,  $|a_n - L| < \epsilon$ "
negates to
"there is an  $\epsilon > 0$  such that for every  $N$  there
is an  $n > N$  with  $|a_n - L| \geq \epsilon$ "

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WHICH IS MECHANICAL. You never have to think about what the negation "means" -- apply the rules, then read the result and see what it says.

Negation is mechanical — flip each quantifier left to right and negate the inside, and never reason about what the negation 'means' until after you have written it down. **Negating correctly is what makes contradiction and contrapositive usable at all** (§§2 and 3), since both techniques begin by negating something, and a wrong negation produces a proof of a different statement.

2 The three techniques

Technique	You assume	Best when
Direct	P, and derive Q.	P gives you something concrete to work with — an equation, a structure, a number.
Contrapositive	not Q, and derive not P.	'not Q' is more concrete than P — see the callout.
Contradiction	P and not Q together, and derive something false.	The claim is a non-existence or an impossibility (§3).
Cases	Each possibility in turn, exhaustively.	The situation genuinely splits — and you must check the cases are exhaustive, which is where this goes wrong.
Counterexample	—	You suspect the statement is false (Module 01 §2).

Try direct first, then contrapositive, then contradiction — in that order, because each is harder to get right than the one before it, and because a direct proof is the easiest for a reader to check. **The ordering is this section's practical content:** most people learning proof reach for contradiction far too early, because it feels powerful and because it always *starts* easily.

CONTRAPOSITIVE IS UNDERUSED, AND IT IS FREQUENTLY THE WHOLE DIFFICULTY

'If n^2 is even then n is even' is awkward **directly** — you know something about n^2 and want a conclusion about n , **which is the wrong direction**: you would have to take a square root and argue about parity.

The contrapositive is 'if n is odd then n^2 is odd' — **and now you can write $n = 2k + 1$, square it, and read off the answer**, which takes one line and no cleverness.

So the rule of thumb: if the hypothesis is hard to use and the negated conclusion is easy to use, take the contrapositive — **which is a mechanical test you can apply before starting** rather than a judgement that comes with experience.

And it is logically identical, not weaker — 'if P then Q ' and 'if not Q then not P ' are the same **statement**, with the same truth value in every case — **so nothing is given up** by the substitution, which is why it is free.

3 Contradiction

- It is the right tool for non-existence — 'there is no largest prime', 'the square root of two is not rational', 'no algorithm decides the halting problem' — because assuming the thing exists hands you an object to work with, and otherwise you have nothing to argue about.
- And for minimal-counterexample arguments: assume the statement fails somewhere, take the smallest failure, and derive a smaller one — **which is induction wearing a different hat** (Module 03 §4).
- But a great many contradiction proofs are direct proofs with an unused assumption — if you never actually used 'not Q' anywhere in reaching the contradiction, then what you proved was Q directly, and you should say so.
- Which is worth checking every single time: find the step where 'not Q' was used, point at it, and if there is no such step, rewrite the proof as a direct one. This is Module 01 §3's where-was-the-hypothesis-used question, turned on your own work.
- And a contradiction proof is harder to read, because the reader has to hold a false assumption in mind throughout and cannot trust any intermediate result — which is a real cost to impose on a reader, and should be paid only when it buys something.

4 Choosing

THE STATEMENT'S SHAPE TELLS YOU WHICH TECHNIQUE TO TRY, WHICH TURNS BEING STUCK INTO A PROCEDURE

'For all x , if P then Q ' → try direct; and if P turns out to be unhelpful, take the contrapositive (§2's callout gives the test).

'There is no x such that...' → contradiction, almost always — because you need the object in hand in order to argue about it, and the assumption is what hands it to you.

'For all $n \geq 1$ ' with n appearing in the statement → induction (Module 03) — **which is the single most reliable signal in this course**: if the statement is indexed by a natural number, induction is the first thing to try.

And 'there exists an x such that...' → construct one, explicitly, or count the possibilities and show the count is non-zero — **which is the existence proof's two flavours**, constructive and non-constructive, and the second is what Module 11 §4's probabilistic method does.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Velleman — How to Prove It, chapters 1 to 3	The whole module — and the quantifier chapter is worth more than its length suggests. Library copy.
Lehman, Leighton & Meyer, chapters 1 and 3 (free)	§§1 to 3 , free — with propositional logic developed enough to make the negation rules obvious.
MIT 6.042J lectures 1 and 2 (free video)	§§2 and 3 — the techniques demonstrated on real statements rather than described.
Hammack — Book of Proof (free)	The whole module, free in full — and it has far more worked examples than this course can include.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Write five statements** from a textbook with their quantifiers made explicit.
- 2 **Find a pair** where swapping two quantifiers changes the meaning, and say how.
- 3 **Negate ten statements** mechanically, including three with nested quantifiers.
- 4 **Negate 'if P then Q'** and explain why the result has no 'if'.
- 5 **Prove three things directly**.
- 6 **Prove three by contrapositive**, and say why direct was awkward.
- 7 **Prove that the square root of two is irrational**.
- 8 **Find a published contradiction proof** that is really a direct proof, and rewrite it.
- 9 **Prove something by cases**, and justify that the cases are exhaustive.
- 10 **For ten statements, name the technique** you would try first, from the shape alone.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What shape do most statements have, and why does that help?
- 2 Why does quantifier order matter, and what is the common error?
- 3 Give the five negation rules.
- 4 Negate 'if P then Q', and explain the result.
- 5 Name the techniques and the order to try them in.
- 6 When is contrapositive the right move?

- 7 Why is it not weaker than a direct proof?
- 8 When does contradiction earn its keep?
- 9 How do you check a contradiction proof is not secretly direct?
- 10 Map four statement shapes onto techniques.